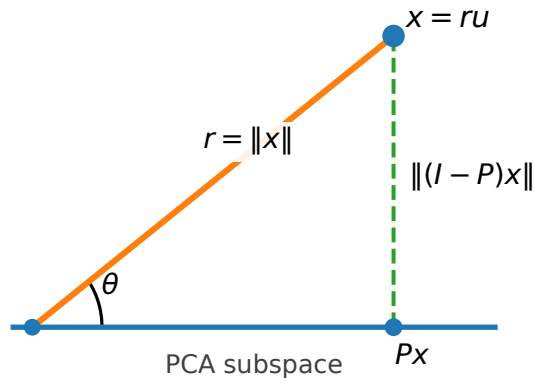


Three geometries of anomaly scoring

(a) Raw distance

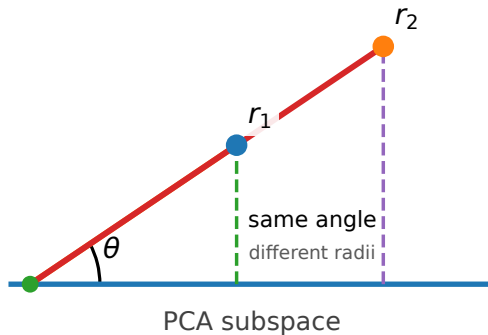
$$D(x) = r^2 \sin^2 \theta$$



Depends on both radius and angular deviation.

(b) Relative angle

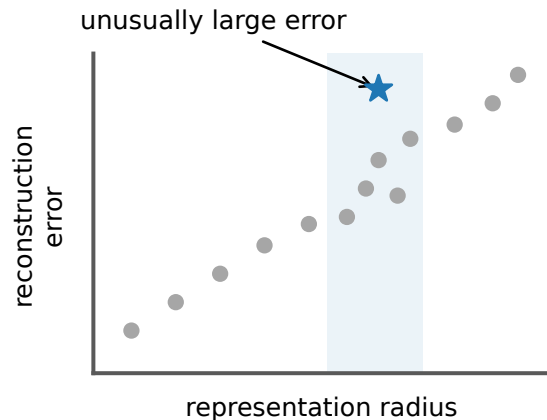
$$R(x) = \frac{D(x)}{\|x\|^2} = \sin^2 \theta$$



Removes radius and keeps angular deviation.

(c) Conditional extremeness

$C(x)$ = local percentile of $D(x)$ among similar r



Ranks residuals among knots with similar radius.